Introduction to Electronic Design Automation

Homework 3

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1 BDD Operation

1. (a)

$$
\begin{aligned}
&\text{robdd\_build}(ab(\neg c + d) + (a\neg b + \neg ab)(c\neg d + \neg cd), 1)\\
&\xrightarrow{\eta} \text{robdd\_build}(b(\neg c + d) + \neg b(c\neg d + \neg cd), 2)\\
&\quad\xrightarrow{\eta} \text{robdd\_build}(\neg c + d, 3)\\
&\quad\quad\xrightarrow{\eta} \text{robdd\_build}(d, 4)\\
&\quad\quad\quad\xrightarrow{\eta} \text{robdd\_build}(1, 5)\\
&\quad\quad\quad\quad v_1\\
&\quad\quad\quad\xrightarrow{\lambda} \text{robdd\_build}(0, 5)\\
&\quad\quad\quad\quad v_0\\
&\quad\quad\quad v_2 = (d, v_1, v_0)\\
&\quad\quad\xrightarrow{\lambda} \text{robdd\_build}(1, 4)\\
&\quad\quad\quad v_1\\
&\quad\quad v_3 = (c, v_2, v_1)\\
&\quad\xrightarrow{\lambda} \text{robdd\_build}(c\neg d + \neg cd, 3)\\
&\quad\quad\xrightarrow{\eta} \text{robdd\_build}(\neg d, 4)\\
&\quad\quad\quad\xrightarrow{\eta} \text{robdd\_build}(0, 5)\\
&\quad\quad\quad\quad v_0\\
&\quad\quad\quad\xrightarrow{\lambda} \text{robdd\_build}(1, 5)\\
&\quad\quad\quad\quad v_1\\
&\quad\quad\quad v_5 = (d, v_0, v_1)\\
&\quad\quad\xrightarrow{\lambda} \text{robdd\_build}(d, 4)\\
&\quad\quad\quad v_2 = (d, v_1, v_0)\\
&\quad\quad v_4 = (c, v_5, v_2)\\
&\quad v_6 = (b, v_3, v_4)\\
&\xrightarrow{\lambda} \text{robdd\_build}(b(c\neg d + \neg cd), 2)\\
&\quad\xrightarrow{\eta} \text{robdd\_build}(c\neg d + \neg cd, 3)\\
&\quad\quad v_4 = (c, v_5, v_2)\\
&\quad\xrightarrow{\lambda} \text{robdd\_build}(0, 3)\\
&\quad\quad v_0\\
&\quad v_7 = (b, v_4, v_0)\\
&v_8 = (a, v_6, v_7)
\end{aligned}
$$

1. (b)

Since

$$f = ab(\neg c + d) + (a\neg b + \neg ab)(c\neg d + \neg cd)$$

we have

$$f_c = abd + (a\neg b + \neg ab)(\neg d)$$
$$f_{\neg c} = ab + (a\neg b + \neg ab)(d)$$

Using procedure `robdd_build` to f_c, we have

$$\begin{aligned}
&\text{robdd_build}(abd + (a\neg b + \neg ab)(\neg d), 1)\\
&\xrightarrow{\eta} \text{robdd_build}(bd + \neg b\neg d, 2)\\
&\qquad \xrightarrow{\eta} \text{robdd_build}(d, 4)\\
&\qquad\qquad v_2 = (d, v_1, v_0)\\
&\qquad \xrightarrow{\lambda} \text{robdd_build}(\neg d, 4)\\
&\qquad\qquad v_5 = (d, v_0, v_1)\\
&\qquad v_9 = (b, v_2, v_5)\\
&\xrightarrow{\eta} \text{robdd_build}(b\neg d, 2)\\
&\qquad \xrightarrow{\eta} \text{robdd_build}(\neg d, 4)\\
&\qquad\qquad v_5 = (d, v_0, v_1)\\
&\qquad \xrightarrow{\lambda} \text{robdd_build}(0, 4)\\
&\qquad\qquad v_0\\
&\qquad v_{10} = (b, v_5, v_0)\\
&v_{11} = (a, v_9, v_{10})
\end{aligned}$$

Using procedure `robdd_build` to $f_{\neg c}$, we have

$$\begin{aligned}
&\text{robdd_build}(ab + (a\neg b + \neg ab)d, 1)\\
&\xrightarrow{\eta} \text{robdd_build}(b + \neg bd, 2)\\
&\qquad \xrightarrow{\eta} \text{robdd_build}(1, 4)\\
&\qquad\qquad v_1\\
&\qquad \xrightarrow{\lambda} \text{robdd_build}(d, 4)\\
&\qquad\qquad v_2 = (d, v_1, v_0)\\
&\qquad v_{12} = (b, v_1, v_2)\\
&\xrightarrow{\lambda} \text{robdd_build}(bd, 2)\\
&\qquad \xrightarrow{\eta} \text{robdd_build}(d, 4)\\
&\qquad\qquad v_2 = (d, v_1, v_0)\\
&\qquad \xrightarrow{\lambda} \text{robdd_build}(0, 4)\\
&\qquad\qquad v_0\\
&\qquad v_{13} = (b, v_2, v_0)\\
&v_{14} = (a, v_{12}, v_{13})
\end{aligned}$$

Hence, the shared ROBDDs of f, f_c and $f_{\neg c}$ are as shown as below (Figure 1).

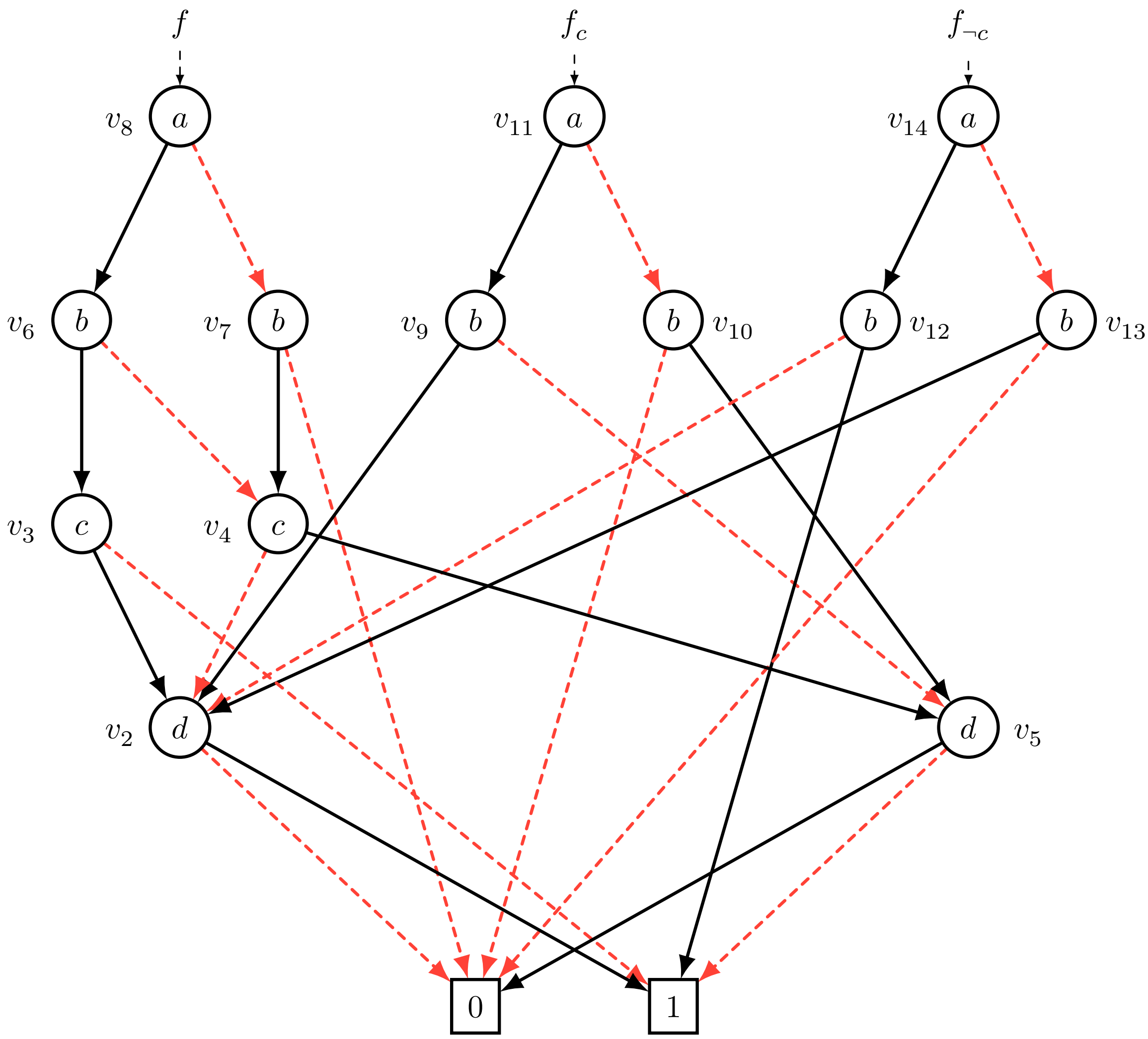

Figure 1: The shared ROBDDs of f, f_c and $f_{\neg c}$.
Red dashed edge represents 0-edge, and black solid edge represents 1-edge.

1. (c)

Since

$$\exists c.f = f_c + f_{\neg c} = \text{ITE}(f_c, 1, f_{\neg c}) = \text{ITE}(v_{11}, 1, v_{14})$$

we can use the ROBDD in Figure 1 to compute the ROBDD of $\text{ITE}(f_c, 1, f_{\neg c})$. Hence we have:

$$\begin{aligned}
&\text{ITE}(f_c, 1, f_{\neg c}) \\
&= \text{ITE}(v_{11}, 1, v_{14}) \\
&= \text{ITE}(a,\ \ \text{ITE}(v_9, 1, v_{12}),\ \ \text{ITE}(v_{10}, 1, v_{13}))) \\
&= \text{ITE}(a,\ \ \text{ITE}(b,\ \ \text{ITE}(v_2, 1, 1),\ \ \text{ITE}(v_5, 1, v_2)),\ \ \text{ITE}(b,\ \ \text{ITE}(v_5, 1, v_2),\ \ \text{ITE}(0, 1, 0)))
\end{aligned}$$

since $v_2 = d$ and $v_5 = \neg d$, we have

$$\begin{aligned}
\text{ITE}(v_2, 1, 1) &= 1 \\
\text{ITE}(v_5, 1, v_2) &= \text{ITE}(\neg d, 1, d) = 1 \\
\text{ITE}(0, 1, 0) &= \text{ITE}(d, 1, 0) = 0.
\end{aligned}$$

Therefore,

$$
\begin{aligned}
&\mathrm{ITE}(f_c, 1, f_{\neg c}) \\
&= \mathrm{ITE}(a,\quad \mathrm{ITE}(b, 1, 1),\quad \mathrm{ITE}(b, 1, 0)) \\
&= \mathrm{ITE}(a, 1, b) \\
&= a + b
\end{aligned}
$$

The ROBDD of $\mathrm{ITE}(f_c, 1, f_{\neg c}) = \mathrm{ITE}(a, 1, b) = a + b$ is as shown as below (Figure 2).

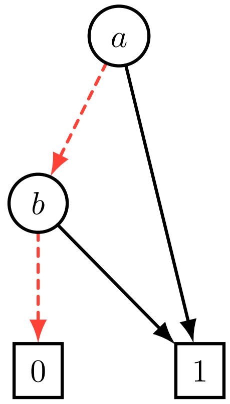

Figure 2: The ROBDD of $\mathrm{ITE}(f_c, 1, f_{\neg c})$.

2 SAT Solving

3 Combinational Equivalence Checking

4 Characteristic Function

4. (a)

The set of states each of which has no direct transition to itself has characteristic function of

$$F(\boldsymbol{s}) = \neg T(\boldsymbol{s}, \boldsymbol{s})$$

4. (b)

The set of states that can be reached from C in exactly two transitions has characteristic function of

$$G(\boldsymbol{s'}) = \exists \boldsymbol{s}, \boldsymbol{s_1}. C(\boldsymbol{s}) \land T(\boldsymbol{s}, \boldsymbol{s_1}) \land T(\boldsymbol{s_1}, \boldsymbol{s'})$$

4. (c)

The set of states that each have a unique next state has characteristic function of

$$H(\boldsymbol{s}) = \exists \boldsymbol{s_1}. (T(\boldsymbol{s}, \boldsymbol{s_1}) \land \forall \boldsymbol{s_2}. ((T(\boldsymbol{s}, \boldsymbol{s_1}) \land T(\boldsymbol{s}, \boldsymbol{s_2})) \rightarrow \boldsymbol{s_1} = \boldsymbol{s_2}))$$

5 Sequential Equivalence Checking